# Driver Overtake Interactions in Formula One

Data Mining (INFSCI 2160)
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## Introduction

Formula One racing overtake interactions can be understood intuitively through the lens of network science. Naturally, drivers can be represented as nodes. Overtake interactions capture the relationships between driver pairs and can be represented as edges in the network. Visualizing and analyzing the network allows for a deeper understanding of driver interactions which is the most interesting and exciting element of racing.

## Overtake Network Visualization

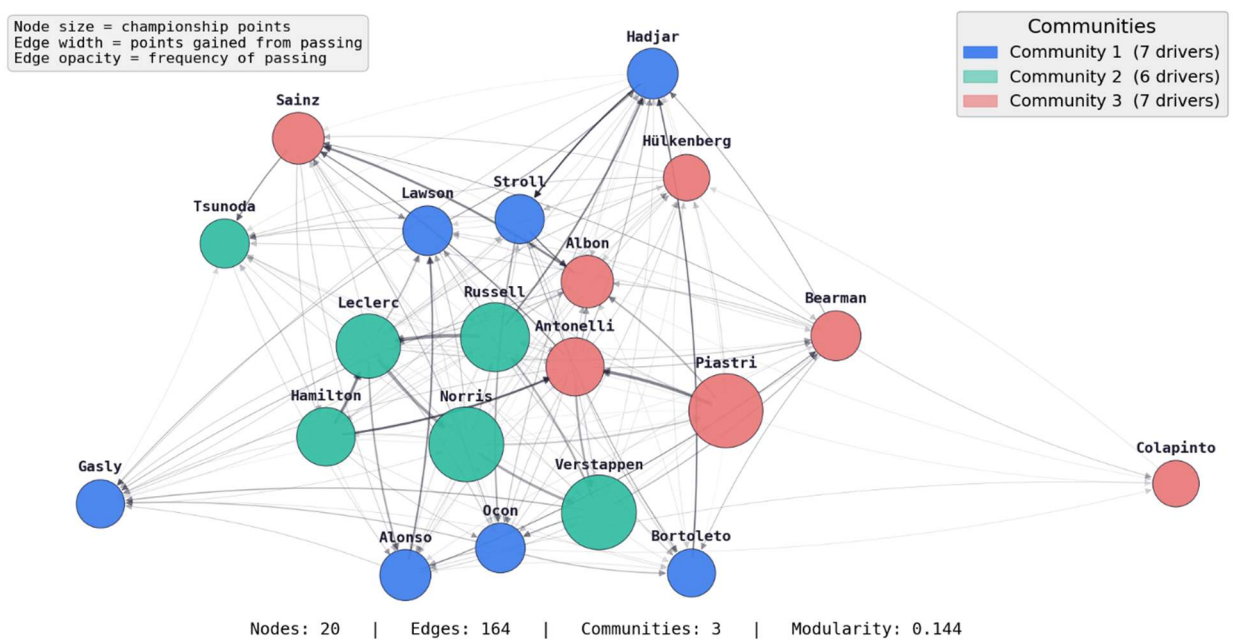


**Figure 1.** F1 2025 Driver Overtake Network. Each node represents a driver; node size reflects final championship points. Node color indicates community membership as determined by the Louvain algorithm. Edge width encodes the net point value gained through passing, and edge opacity encodes the frequency of passing between each pair of drivers. Directed edges point from the net overtaker to the net overtaken driver.

## Results

- The network is fully connected. All 20 drivers have a net dominant passing direction against at least one other driver across the season.
- An average clustering coefficient of 0.905 and average path length of 1.59 confirm strong small-world behavior. The nodes have an average unweighted degree of 8. Drivers who share common opponents tend to also interact directly with each other; however, most of the drivers also interact with nearly half of the rest of the field. Both of these in combination imply that the sport is competitive and drivers are interacting all over the field over the course of the season.
- The thickest edges connect the championship contenders, reflecting that position changes higher up the grid carry the most significance. Drivers towards the back of the field reflect limited valuable interactions in terms of points. Colapinto and Gasly are pulled to the layout periphery in the visualization which is consistent with their low point totals and minimal high-value passing interactions.
- Three driver communities were detected using the Louvain algorithm. Low modularity (0.144) indicates weak community boundaries which means communities are suggestive rather than definitive. The three communities loosely correspond to championship tier (see figure 2):
    - *Community 2* captures the front-runners
    - *Community 3* is mostly the midfield
    - *Community 1* includes drivers towards the back of the field

| Driver | Total Points | Community |
|---|---|---|
| Lando Norris | 423 | 2 |
| Max Verstappen | 421 | 2 |
| Oscar Piastri | 410 | 3 |
| George Russell | 319 | 2 |
| Charles Leclerc | 242 | 2 |
| Lewis Hamilton | 156 | 2 |
| Kimi Antonelli | 150 | 3 |
| Alexander Albon | 73 | 3 |
| Carlos Sainz | 64 | 3 |
| Fernando Alonso | 56 | 1 |
| Nico Hulkenberg | 51 | 3 |
| Isack Hadjar | 51 | 1 |
| Oliver Bearman | 41 | 3 |
| Liam Lawson | 38 | 1 |
| Esteban Ocon | 38 | 2 |
| Lance Stroll | 33 | 1 |
| Yuki Tsunoda | 33 | 1 |
| Pierre Gasly | 22 | 1 |
| Gabriel Bortoleto | 19 | 1 |
| Franco Colapinto | 0 | 3 |

**Figure 2.** F1 2025 Final Championship Standings & Louvain Communities.

## Significance:

Formula One is typically analyzed through individual performance metrics such as lap times, pit stop strategy, and championship points. This project takes a different approach by modeling the 2025 season as a directed weighted network of driver interactions, where edge weights reflect the championship significance of each overtake rather than a raw count of overtakes. This framing reveals competitive insights including the small-world clustering of the field, the degree to which community structure aligns with championship tier, and the relative isolation of drivers with limited high-value interactions. The methodology can be applied to other seasons or periods of time (e.g. regulation eras).

## Data Overview

The data that was analyzed for this project is available on Kaggle. The information provided in the dataset is clear, accurate, and well-organized. Comprehensive and usable lap-by-lap data including racing positions for each driver is available from 1996-present. The data reported here includes the full 2025 season comprised of 24 grand prix races & six sprint (mini) races.

## Data Preparation & Network Construction

Drivers with fewer than 10 total races were removed from the dataset entirely. If a driver was disqualified from a race, then that driver was removed from the dataset for that particular race. Overtakes were identified by iterating through each driver's position on a lap-by-lap basis for each race of the 2025 season. A position gain of one or more places relative to the previous lap was recorded as one overtake interaction per position gained. For each individual overtake, the displaced driver was identified using a lap-position lookup table built during the process of constructing the main overtake dataset. The overtake dataset was built using the original raw data. Any passes that were identified as self-overtakes due to data inconsistencies were removed. 3,766 overtakes were identified and analyzed.

The overtakes were weighted to reflect the Formula 1 point system (25-18-15-12-10-8-6-4-2-1 for grand prix positions 1-10, and 8-7-6-5-4-3-2-1 for sprint race positions 1-8). Overtakes occurring outside of the points (i.e. for positions 11 or higher in grand prix races and 9 or higher in sprint races) received a value of zero. This weighting scheme captures the significance of each overtake rather than treating all passes equally. The direction and weight values of the edges reflect net interactions between each driver pair. The edge direction always points from the net overtaker to the net overtaken driver. Edges cannot be bidirectional by definition. There were ultimately 164 edges in the network.

The network was visualized using a force-directed spring layout (k=7.5, 200 iterations) and the node sizes are scaled to reflect the final 2025 championship points (larger node implies more points). The communities are easily identifiable in the network visualization via color. Edge width reflects points gained from passing. Edge opacity reflects frequency of passing. All code was written in Python using pandas, NetworkX, and matplotlib.

**Github Repository:** The code, datasets, slides, and this report are all archived at: https://github.com/mnh22/Formula-One-Overtake-Network